TOPIC 25/60

What is PEFT?

Parameter-Efficient Fine-Tuning.

How to train massive models on tiny consumer GPUs.

Ashish Jangra
linkedin.com/in/ashish-jangra

THE PROBLEM

Fine-tuning a 70-billion parameter model normally means updating 70 billion weights. This requires massive server clusters costing thousands of dollars.

The Bottleneck:

Traditional fine-tuning isn't just expensive; it's physically impossible for individual developers to run on home or laptop hardware.

Ashish Jangra
linkedin.com/in/ashish-jangra

THE PEFT SHIFT

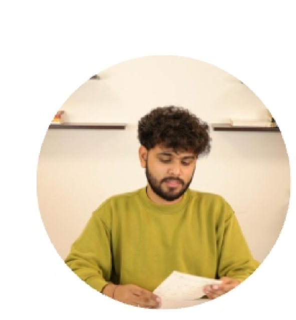
🤏

Less is More

PEFT assumes you don't need to change every single brain cell to learn a new task. You only need to adjust a tiny, targeted subset.

The Strategy: "Freeze" 99.9% of the model's weights and only train the tiny remaining fraction.

Ashish Jangra
linkedin.com/in/ashish-jangra



THE TECHNIQUE

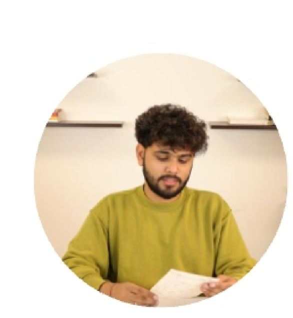
LoRA

Low-Rank Adaptation

Think of the AI's weight matrix as a giant 1000x1000 grid. Instead of training it, we add a tiny, lightweight "adapter" layer on the side.

> We only train these thin adapter layers. At inference time, we just "overlay" these weights on top of the original model.

Ashish Jangra
linkedin.com/in/ashish-jangra



THE MATH

Intrinsic Dimension

Research suggests that even giant models have a low "intrinsic dimension"—meaning they don't actually need all those parameters to learn a new task.

> By focusing only on these low-rank adaptations, we reduce memory usage by **1,000x** while keeping almost all the performance.

Ashish Jangra
linkedin.com/in/ashish-jangra

THE BENEFITS

Why use PEFT?

PEFT isn't just about saving money; it changes the AI development workflow entirely.

- **Hardware:** You can fine-tune LLMs on a single GPU.
- **Storage:** An adapter file is only 10MB, compared to the 100GB+ full base model.
- **Modularity:** You can swap LoRA adapters in and out for different tasks instantly.

Ashish Jangra
linkedin.com/in/ashish-jangra

COMMON TECHNIQUES

LORA

The Industry Standard

Adds tiny trainable layers to the linear weights. Most widely used.

PREFIX TUNING

Adding Trainable Vectors

Instead of changing weights, we prepend learnable "prompt vectors" to the model's inputs.

Ashish Jangra
linkedin.com/in/ashish-jangra

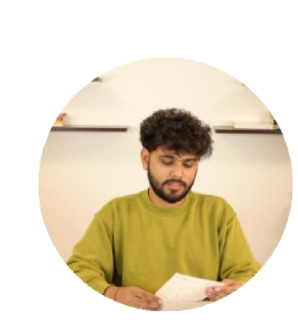
DEPLOYMENT

Plug & Play

Because adapters are small, you can host 1 massive "Base Model" and load thousands of tiny "LoRA Adapters" on top for different users.

Imagine a platform where one base AI becomes a *Lawyer*, a *Coder*, or a *Poet* just by switching a tiny 10MB adapter file.

Ashish Jangra
linkedin.com/in/ashish-jangra



FULL FT VS PEFT

Full Fine-Tuning

Requires massive compute, stores a new copy for every task, prone to catastrophic forgetting.

PEFT (LoRA)

Works on consumer hardware, incredibly portable adapters, retains base model integrity.

Ashish Jangra
linkedin.com/in/ashish-jangra

PEFT (LoRA)

CONCEPT UNLOCKED

- ⚡ Cheap & Fast Training
- 📉 Tiny Model Files
- 🧩 Modular Adapters
- 🏠 Works on Home GPUs

CONNECT WITH ME

LinkedIn

GitHub

Kaggle

Instagram

Ashish Jangra

linkedin.com/in/ashish-jangra